

4-1-1: Examining network Traffic Issues

At the end of this episode, I will be able to:

1. Identify and explain common network traffic issues (including adverse effects), given a scenario.

Learner Objective: *Identify and explain common network traffic issues (including adverse effects), given a scenario.*

Description: In this episode, the learner will examine common issues with network traffic that can hinder performance. We will explore various conditions such as congestion, jitter, packet loss, bandwidth issues, and more.

- Congestion and contention
 - o When network demands exceed capacity, causing a traffic jam of data.
 - o Effect on performance
 - ♣ Leads to slower performance
 - ♣ Increased latency
 - ♣ Timeouts
 - o Solution examples
 - ♣ Increase bandwidth
 - ♣ Implement Quality of Service (QoS) rules and traffic prioritization.
- Bottlenecking
 - o A point in the network that significantly reduces performance due to limited capacity.
 - o Effect on performance
 - ♣ Causes slow data transfer rates and long wait times.
 - o Solution example
 - ♣ Upgrade hardware
 - ♣ Increase link capacity
- Bandwidth
 - o The maximum data transfer rate of a network connection.
 - o Effect- Insufficient bandwidth results in slow data transfers and poor network performance.
 - o Solution
 - ♣ Upgrade network infrastructure for higher bandwidth capabilities
- Throughput capacity
 - o The actual rate at which data is successfully transferred through the network.
 - o Effect on performance
 - ♣ Often lower than bandwidth due to factors like latency and packet loss, affecting performance.
 - o Solution example
 - ♣ Upgrade network components for higher throughput
- Latency
 - o The time delay from the moment information is sent to when it's received.
 - o Effect on performance
 - ♣ Can significantly impact time-sensitive applications.
 - o Solution
 - ♣ Optimize routing, reduce network hops
 - ♣ CDNs for better distribution
 - ♣ Implement Quality of Service (QoS) rules and traffic prioritization.
- Packet Loss
 - o The failure of one or more data packets to arrive at their intended destination.
 - o

- Effect on performance

- ♣ Can result in increased retransmissions
- ♣ Reduced network throughput.

o Solution

- ♣ Troubleshoot and correct
 - ♣ Network errors
 - ♣ Replace physical connections
 - ♣ Implement Quality of Service (QoS) rules and traffic prioritization.

- Jitter

- o The variation in the delivery time between packets arriving caused by network congestion, timing drift, or route changes.

- o Effect on performance

- ♣ Problematic for real-time communications, leading to poor audio and video quality.

- o Solution

- ♣ Use jitter buffers and traffic prioritization to minimize impacts.

- Additional Resources:

- o If applicable